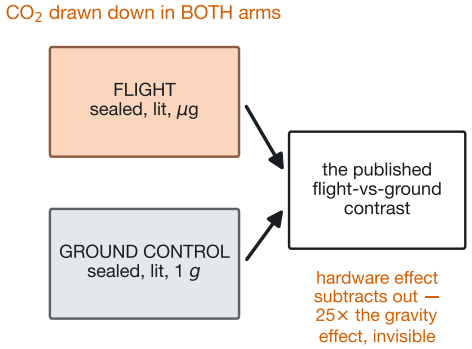
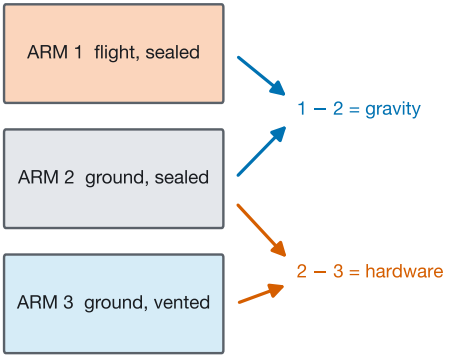
# Designing the experiment that could tell hardware from gravity

## a What the standard contrast cannot see




What survives is a persistent ~5–9 % assimilation penalty at saturating light (under 3 % at OSD-522's), and downstream of it a carbon-starvation signature — which is what the omics show.

## b The design that separates them




Two arms confound the two effects. Three separate them — and the third arm needs no flight.

## c The gap this ran into

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OSDR metabolomics studies that are of a plant

**12 of 567**

OSDR studies that are metabolite profiling, all mouse, human, rat or microbial

**0 known**

in-canister $CO_2$ logs, any BRIC-class flight

A ~20-compound targeted panel on archived material would be the first plant spaceflight metabolome.